The vertex / MPM interface: particle-to-surface contact, and the spheroid it was built for

`prototype/ecm/test_03_mesh_contact.py` · `test_04_spheroid_ecm.py` · `log/okuda_ECM/03*, 04_spheroid_ecm` · 10 August 2026

1. The problem, and why the obvious scheme does not fit

The epithelium is a triangulated surface with no mass; the matrix and the basement membrane are MPM continua on a background grid. Every mechanism between them — adhesion, contact, load transmission — has to cross that boundary, and in this prototype it never has: the epithelium is a recorded surface that pushes and is never pushed back.

Outside biology the problem is standard, and it is called FEM − MPM coupling. The natural scheme, *grid-node* coupling (CFEMP, Lian et al. 2011), resolves contact by comparing the two bodies' velocities at grid nodes they share, and it requires the mesh and the grid to be *comparable in size*. Ours are not. A cell is $0.73\,\Delta x$ and the basement membrane $0.1\,\Delta x$, so both bodies fall inside one cell, the grid hands them a single mass-weighted velocity, and they are welded: this is exactly what runs 142–151 measured ten times over. *Particle-to-surface* coupling (ICFEMP, Chen et al. 2015) removes the restriction by testing a particle against a mesh *face* rather than against a node, and that is what is implemented here.

2. The interface, in equations

For particle p under face f with outward normal $\hat{\mathbf{n}}$, let $\mathbf{w}$ be the barycentric weights of p's projection on f and $z_f = \sum_i w_i z_i$ the interpolated surface. The penetration, the normal penalty, and the regularised Coulomb friction are

$$d_p = \big[(\mathbf{x}_p - \mathbf{x}_f)\cdot\hat{\mathbf{n}}\big]_+, \qquad \mathbf{f}_p^{\,n} = -\,k\,d_p\,\hat{\mathbf{n}}, \tag{1}$$

$$\Delta\mathbf{v}_t = \big(\mathbf{v}_p - \textstyle\sum_i w_i\mathbf{v}_i\big)_{\perp\hat{\mathbf{n}}}, \qquad \mathbf{f}_p^{\,t} = -\min\!\Big(\mu\,kd_p,\ \mu\,kd_p\tfrac{|\Delta\mathbf{v}_t|}{\varepsilon}\Big)\frac{\Delta\mathbf{v}_t}{|\Delta\mathbf{v}_t|}, \tag{2}$$

and the reaction is distributed to the face's vertices by the same weights, which is what makes the exchange conservative by construction:

$$\mathbf{f}_i \;=\; -\,w_i\left(\mathbf{f}_p^{\,n} + \mathbf{f}_p^{\,t}\right), \qquad \text{so} \qquad \sum_p \mathbf{f}_p \;+\; \sum_i \mathbf{f}_i \;=\; \mathbf{0}. \tag{3}$$

ε is a slip velocity, not a fudge: it is the speed below which a contact is treated as stuck, so a resting contact does not chatter.

The one constraint that is not optional. The contact force reaches the particle as a body force, integrated at Δt, so it is stable only while $\Delta t\sqrt{k/m} < 1$, i.e.

$$k \;<\; k_{\max} = \frac{m}{\Delta t^2}, \qquad m = \rho\,\Delta x^3/\mathrm{ppc} = 9.5\times10^{-7}, \quad k_{\max} = 95. \tag{4}$$

The first version of this rig used $k = 3\times10^4$, three hundred times past it. The stiffness is therefore written as a *fraction* of $k_{\max}$ — here 5%, so $k = 0.238$ — which is the same discipline the flat sheet needed after it NaN'd at $k = 6\times10^3$.

3. As a Plexus2 container

The contact is a *relation*, so it is an edge set and not a force field on a body:

object	kind	carries
`contact`	edge set, `pre: mpm_particle`, `post: vertex`	d_p, $\mathbf{w}$, bound flag
`mesh_contact`	**Lateral** on `contact`	returns a delta to *both* endpoints
`contact_rebuild`	**Rewire** on `contact`	which particle is under which face

Two placements matter. `mesh_contact` returns `{mpm_particle: f_p, vertex: f_i}` from one call, so Eq. (3) cannot be half-implemented — which is precisely how the spheroid's adhesion lost its reaction. And it belongs *inside* the `substep_dt` block, recomputed at the substep positions: at frame level a stiff contact is integrated at Δt_{frame} and the ceiling in Eq. (4) falls by $(\Delta t_{\text{frame}}/\Delta t_{\text{sub}})^2 = 400$.

4. The rig, and what it measures

A flat patch of 17×17 vertices with four-neighbour springs, pressed at a prescribed speed into a block of 81,202 MPM particles; mesh cell $1.68\,\Delta x$, no gravity. Three variants, because one of them alone cannot separate the interface from its boundary conditions: the block free, the block trapped against a rigid floor, and the same with a hole cut in the surface. The floor is a *grid* boundary condition — the downward component of the grid velocity is zeroed below the plane, which is how MPM states an obstacle — and the hole is a real hole: 21 vertices inside $r = 0.06$ are removed, the 52 springs touching them go with them, and every quad with a missing corner stops being a contact face, so the material under it has nowhere to go but through.

run	momentum residual	penetration	slip, $\mu = 0.4$ / $\mu = 0$	compression
`03` free block	1.5×10^{-7}	0.67 cells	—	—
`03b` rigid floor	1.6×10^{-7}	1.47 cells	0.069 / 0.368	15.0%
`03c` floor + hole	1.4×10^{-7}	1.56 cells	0.088 / 0.344	14.4%

The momentum residual is float32 machine precision in all three, so the reaction of Eq. (3) is really returned — the test a one-way coupling fails by construction, and the one the spheroid's adhesion would fail today. **Friction changes the slip four- to fivefold**, so the contact is not MPM's automatic weld wearing a coefficient; the free-block run cannot say this, because its slip is measured over the frames that have a contact and it lifts off before enough of them accumulate. **Penetration is bounded but is one to one and a half cells**, which is a lot: the fix is a stiffer penalty with a smaller substep, or barrier contact. And the floor is what turns an indentation into a confined compression — with nowhere to go the block loses 15% of its height, and the hole gives that back slightly (14.4%) because material escapes through it.

The particles are coloured by $|J - 1|$, the volumetric strain, on one 99th-percentile scale taken over the whole run. That scale cannot be known before the run, so the renderer steps the physics once, keeps the drawn frames, and sets the scale from them — the alternative, a value guessed in advance, is what made the falling-block movie a solid saturated colour.

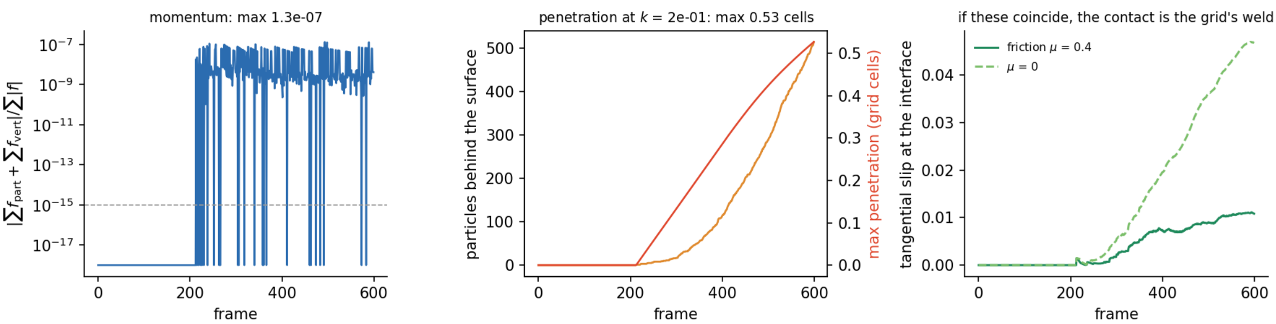

Figure 1: Run `03`. Left: the momentum residual, which is the test that a one-way coupling fails by construction. Middle: how many particles are behind the surface and how deep, in grid cells. Right: tangential slip with friction and without — if these two coincided, the "contact" would be MPM's automatic weld wearing a friction coefficient.

Two failures the rig produced before it produced a result, both worth keeping. Under gravity the block *fell out from under the patch* — its floor is twenty cells below its underside, so it was in free fall and the two bodies never met (block top $0.5800 \to 0.5728$ while the patch went $0.5850 \to 0.5778$). And a patch of independent vertices is not a mesh: pressing its rim left the interior behind, so the lattice needed edge springs before a press could load anything.

5. What this does and does not license

It licenses the interface: a surface and a continuum can exchange force conservatively at a resolution where grid-node coupling cannot, with slip that is a material property rather than a numerical accident, and it survives an obstacle behind the material and a hole in the surface without either being special-cased. It does not license the geometry — the patch is flat and axis-aligned, so face lookup is $O(1)$ and a real mesh needs a BVH — and it does not license the accuracy: half a grid cell of penetration is a lot, and the fix is either a stiffer penalty with a smaller substep or the barrier formulation (BFEMP, Li et al. 2022), which forbids penetration outright at the cost of an implicit solve. The next question is whether the same operator survives a *curved, moving* mesh, which is the spheroid.

6. The spec this argues for

Everything above, plus the archived line and the literature the note beside this one records, points at one container. It is chosen so that the *expensive* fidelity is the fidelity we cannot do without, and the cheap fidelity is taken in full.

What resolution cannot buy. A basement membrane is 100 nm on a tissue of 100 μm. To carry it as a resolved MPM body – seven cells through its thickness, which is what the pillar rig needed – costs $\Delta x \approx 3 \times 10^{-4}$ and a grid of 3500^3: four times 10^{10} cells, for one sheet. So the sheet is *codimensional*: its thickness enters as a parameter (a rest length, Eq. **??**), never as resolved geometry, which is the move thin-shell MPM and C-IPC both make. For the same reason its adhesion is an explicit relation and not grid contact: below a cell the grid welds, and runs 142–151 measured that ten times.

	representation	why not finer	cost
epithelium	vertex model, replayed	pass 1 and pass 2 cannot share a world	already paid
basement membrane	MPM particles, codim-1, thickness as ℓ_0	3500^3 grid to resolve it	45k particles
matrix	MPM fibres, $\Delta x = 1/64$	the stress front is resolved at 1/64	140k particles
plaque	edge set `cell`$\to$`bm`, $\Sigma \approx 7T$	resolved plaques need the same 3500^3	4–20k edges
fibril	edge set `bm`$\to$`ecm`	—	4k edges

The schedule, and the one placement that matters. `plaque_pull` and `fibril_pull` go *inside* the substep block, with the MPM cycle; at frame level a stiff relation is integrated at Δt_{frame} and its stability ceiling is 400× lower, which is the whole difference between run 121 and run 133. The mass balance and the proteolysis stay at frame level, where their timescale is hours:

$$\text{aggregate} \to \text{seed} \to \Big[\underbrace{\text{plaque\_pull},\ \text{fibril\_pull},\ \text{mpm\_strain},\ \text{scatter},\ \text{grid\_update},\ \text{gather}}_{\Delta t_{\text{sub}}=2\times 10^{-4}}\Big] \to \text{bm\_secrete},\ \text{bm\_degrade},\ \text{e}$$

The fidelity that is nearly free, and is therefore not optional. A per-particle areal density with production and removal (Eq. **??**) at the measured half-life of 3–10 hours — one scalar per particle, and the mechanism without which a sheet cannot enclose a tripling radius at all, as run 155 showed elasticity alone cannot. A strain-dependent modulus (Eq. **??**), one line in the constitutive law, against a sheet that reaches $\lambda = 2.4$. A stiffness ratio of 10–100 between sheet and stroma, which costs nothing but a second parent set. And punctate adhesion at the measured spacing, which is an edge list rather than a field.

What it costs and what it still does not buy. The added operators are $O(N)$ scalars and two edge lists, so a 402-frame run stays where the current line is: roughly 45 minutes on one A6000 at 185k particles and $n_{\text{grid}} = 64$. What it does not buy is two-way mechanics with the epithelium — the tissue is still a replay, so it drives and is never driven. The operator in this note is exactly the piece that removes that limitation, and the next experiment is the one that puts a *curved, moving* mesh on the same contact: the spheroid.

7. The curved, moving mesh: `04_spheroid_ecm`

That experiment is now run. It combines the three pieces the neighbouring notes settled — the tissue of `01c` (two-pool myosin, cytokinetic ring at 1), the matrix of `02h` (20 particles per strand, eight substeps, `drag`= 8, `store_stress` with `measure: vonmises`), and the contact of §2 above — and it deletes the two operators that were standing in for the interface: `cell_to_ecm[replay]`, a radial force read off a smoothed 32×64 radius map, and `cell_exclude_3d`, the projection that swept up whatever the penalty failed to hold. The first cannot return a reaction (a bin of a map has no vertices) and cannot generate a shear (a radial force has no tangential part); the second makes non-penetration unmeasurable, because with it on the answer is zero by construction.

The lookup, which is the only thing the curved surface needed. The tissue is star-shaped about its own centroid, so a direction names the face — the bin structure `apical_map` already uses, holding the *triangles* instead of a radius. Two details are not free. The grid is re-sized every frame, because a nine-bin query is exact only while a triangle fits in a bin and the faces shrink from 0.17 to 0.028 rad as the tissue divides: 82 bins in 8 rows at frame 0, 3576 in 53 rows at the last, at most 64 triangles in a bucket. And the rows are *reduced* — each given as many ϕ bins as it can hold square ones — because on a plain (θ, ϕ) lattice a bin near the pole is a sliver, a triangle spans ten of them, and the query misses the face the particle is standing on: silently, only near the poles, with matrix inside the tissue as the symptom. `selftest()` certifies the whole lookup against a test on *every* triangle before any run uses it: coverage 1.00000 and zero disagreement out of 400 rays, on the meshes of frames 0, 100 and 199.

The regime, in numbers, since it is the reason the scheme has to be this one. The spheroid is scaled so its final radius is 0.15 box units, 9.6 grid cells; a cell face is then 0.27 to 0.42 Δx over the whole run. Both bodies

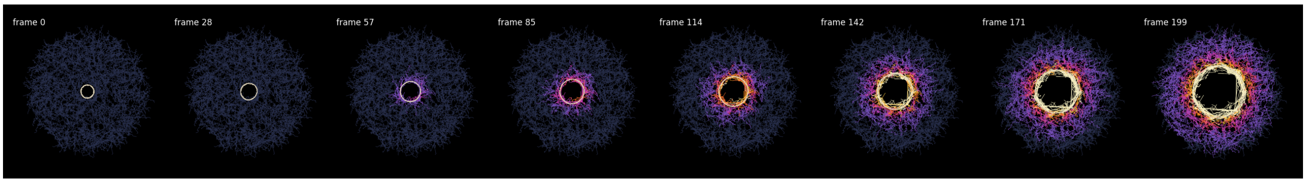

Figure 2: `04`, the section, coloured by von Mises stress on one scale over the whole run (full scale 7.75, the p99). The spheroid clears a hole in the matrix, and a shell of compacted, stressed strands travels outward ahead of it while the far field stays unloaded. The matrix nearest the tissue reaches a density 1.24 times its own seeded value at $r = 0.158$, just outside the surface.

live inside one grid cell, which is where grid-node coupling welds them — the thing runs 142–151 measured ten times over.

what	`03` flat patch	`04` spheroid	
momentum residual	1.5×10^{-7}	1.2×10^{-7} max, 1.5×10^{-8} median	float32 precision, so the reaction is real
penetration	0.67–1.56 cells	0.82 cells	
particles behind the surface	—	4183 of 200,000	0.90 per particle *in contact*
first contact	prescribed	frame 16	the cavity is 0.045 and the tissue reaches it ther
solve	—	209 s, 199 frames	200k particles, 8 substeps, one A6000

The particles behind the surface are the contact, not a leak, and that is a measurement rather than an assurance: their count stays at 0.90–0.96 of the number of particles in contact for the whole run and their greatest depth is 0.77 cells. A penalty holds a particle where its own push balances the surface's advance, so a layer one penetration-depth thick is what a penalty *is*; a leak would be a ratio that climbs. This is the number `cell_exclude_3d` was hiding.

The matrix responds as an isotropic elastic shell, which is the right answer and not a disappointing one. The radial displacement, binned by the particle's own initial radius so a bin is a material shell rather than a region of space, is fitted against the closed form for a spherical shell driven at the inside with a free outer surface, $u = C_1 r + C_2/r^2$: $R^2 = 0.967$, 0.999, 0.988, 0.943 at frames 50, 100, 150, 199. The local slope is -1.03 against the shell's own -1.26 — and neither is the -2 of an infinite incompressible continuum, because a free outer surface at 2.7 tissue radii adds the $C_1 r$ term. Quoting r^{-2} against this geometry would be comparing a bounded shell with an unbounded medium; quoting $r^{-0.5}$ to r^{-1} as evidence of fibres [1] would be reading that same boundary as a mechanism.

What the spheroid does to the strands, which is what the run was to check. Everything depends on where the strand started, and a tissue-wide mean is the far field's answer wearing the whole matrix's name — the mean end-to-end stretch is 1.031, and the shell nearest the tissue is at 1.81:

the radius the strand started at	0.052	0.070	0.090	0.115	0.150	0.320
end-to-end stretch	1.812	1.566	1.306	1.138	1.042	1.013
bow, over its own frame 0	3.432	2.708	1.861	1.412	1.154	1.010
$\langle \cos^2 \theta \rangle$ to the radius, measured	0.055	0.066	0.087	0.130	0.213	0.311
the same, affine control	—	0.016	0.040	0.112	0.211	0.310

And the turning is affine, which is the control that makes the number mean something. The strands end up strongly circumferential, and a matrix pushed outward does that to its own material lines whether or not it is fibrous. So the frame-0 strands are carried through the run's own measured radial displacement profile — spherically symmetric by construction, no mechanics in it — and read the same way. Beyond 1.5 tissue radii the two agree to the third decimal (0.213 against 0.211, 0.311 against 0.310); over the whole inner shell, 0.2725 measured against 0.2714 affine. The excess is entirely inside 0.1, where the measured strands are *less* circumferential than affine (0.066 against 0.016), which is rearrangement against a bumpy moving surface and not a fibre effect — the constitutive law reads the strand direction nowhere, exactly as `note_fibre` measured. Anything else here would have been a defect.

And the reaction now exists as a force on named vertices. It is recorded per frame, and also binned by direction and divided by the area each bin covers, which is the pressure map `ecm_growth_gate_3d` reads. The clamp this operator applies to its own force before recording it — so that the momentum test cannot pass on a force the scatter later clipped — never bound: the largest contact acceleration over the run is 1.8×10^3 against a clamp of 3×10^4 and a stability ceiling of 6.25×10^6.

What this licenses and what it does not. It licenses the interface on the geometry it was built for: a curved,

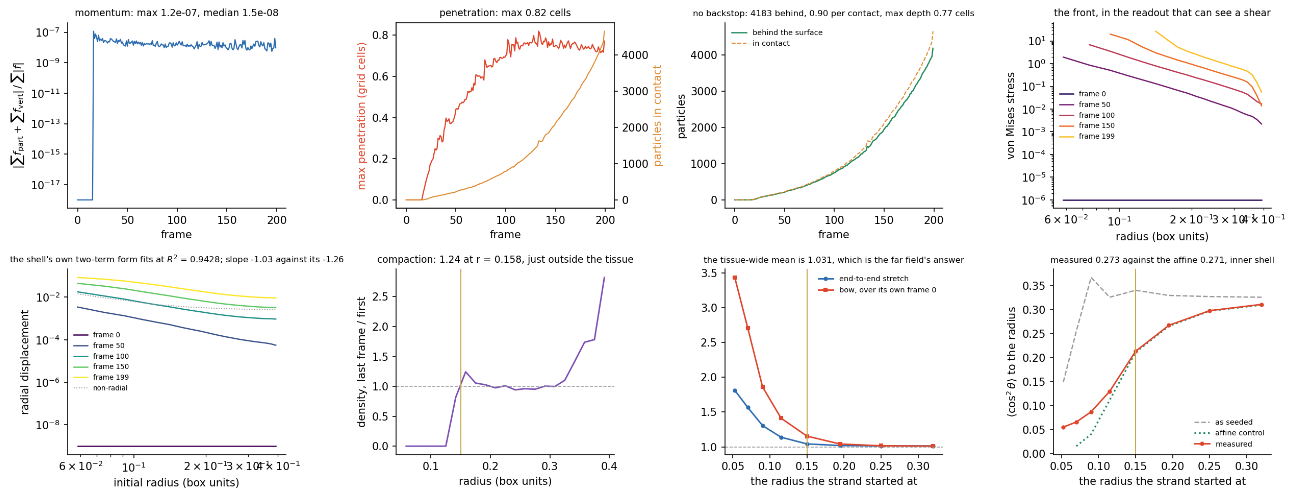

Figure 3: `04`. Top: the three interface measurements — the momentum residual, the penetration, and the particles behind the surface against the particles in contact, which is what says they are the contact layer rather than a leak — and the stress front. Bottom: the radial displacement against the shell's own two-term form; the compaction, with the tissue's final radius marked; the fibre state against the radius the strand started at; and the reorientation against its affine control, which the outer bins sit on exactly.

dividing, re-meshed surface exchanges force with an MPM continuum conservatively, at a resolution three times finer than the grid, with the non-penetration it fails to achieve *measured* rather than projected away. It does not license two-way mechanics — the tissue is still a replay, and the reaction it now computes is recorded and dropped; handing `vertex_force` to a live epithelium is the next run and is a schedule change rather than an architecture change. It does not license the far field, which ends at 2.7 tissue radii against a free surface. And it does not license anything fibrous: every number above is the isotropic continuum's, which is what `note_fibre` predicted and what the anisotropic term it proposes has to beat.

References

[1] Wang, H., Abhilash, A.S., Chen, C.S., Wells, R.G., Shenoy, V.B. (2014) Long-range force transmission in fibrous matrices enabled by tension-driven alignment of fibers. *Biophys. J.* **107**(11):2592–2603.